# Introduction

In recent years, generative AI (Gen AI) and large language models (LLMs) have reshaped the boundaries of what software can do. From chatbots that sound eerily human to code-generation tools that augment developer productivity, the rise of transformer-based architectures has opened up a new paradigm in application design. But while much of this innovation has played out in Python-centric ecosystems, Java developers—especially those working on enterprise-scale systems—have often found themselves on the sidelines, grappling with integration gaps, unclear tooling, or limited examples.

This book aims to change that.

***Generative AI-Driven Application Development with Java*** is a hands-on, code-first guide built explicitly for Java practitioners. Whether you're a backend engineer fluent in Spring Boot or a performance-minded developer building reactive systems in Quarkus, this book provides a clear, structured roadmap for integrating LLMs into modern Java applications.

You'll start from first principles, understanding what makes Gen AI "generative," how LLMs like GPT-4o and Mistral work, and why models are only one part of a much larger ecosystem. From there, you'll dive into building real-world applications such as chat assistants, retrieval-augmented systems, tool-using agents, and even multimodal apps that read and write images. You'll explore hosted APIs from OpenAI and Azure and also self-host your own models using tools like Ollama and Jlama—all without leaving the JVM.

But this book goes beyond the "Hello, World" demos.

With each chapter, you'll learn to wire up production-grade systems: prompt engineering patterns, memory management, evaluation harnesses, API guardrails, and observability pipelines. You'll explore the new breed of Java-native AI tooling—from LangChain4j and Spring AI to DJL and ONNX—while keeping scalability, security, and maintainability front and center.

This book is not just for Java developers looking to "add a chatbot." It's for engineers, architects, and DevOps professionals who need to bring AI into their core platforms—re-

sponsibly, reliably, and at scale.

## Who Should Read This Book

**This book is written for:**

- Java developers and architects eager to add LLM capabilities to new or legacy systems
- Backend engineers seeking a code-first approach to integrating GenAI
- DevOps and MLOps professionals looking to operationalize AI features in production
- Educators and data scientists who prefer Java-centric workflows over Python

No prior experience with AI or machine learning is required, but a working knowledge of Java and modern application development frameworks (like Spring Boot or Quarkus) will help you get the most out of the examples and exercises.

By the time you finish this book, you won't just understand how to connect Java to LLMs—you'll know how to **build**, **test**, **observe**, and **ship** real-world AI applications with confidence.

Let's get started.

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